

Rahul Rawat

Master Thesis Agentic Pentesting and AI Agents for Pentest Workflows

PERSONAL DETAILS

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Portfolio	rah-portfolio.pages.dev
Date of birth	20 February 2002
Nationality	Indian, student visa with valid work permit
Availability	Werkstudent 20 hours per week immediately, full time from April 2027

PROFILE

Data Science and Analytics Master's student at SRH Heidelberg based in Mannheim with hands on practice in AI agents, Retrieval Augmented Generation and LLM orchestration. I built a Hybrid RAG Orchestrator with agentic routing on top of Llama **3.1 8b** via Groq and LangChain, delivered a Fairness by Design credit scoring system with honest evaluation aligned to the EU AI Act, and shipped an end to end time series pipeline at eRay GmbH with gate checks and governance rules. Comfortable in Python, LangChain, Git and API based LLM services, with a clear eye for evaluation, guardrails and traceable agent behaviour.

EXPERIENCE

Oct 2025 to Mar 2026
Heidelberg

eRay GmbH
Data Scientist

- In a **6 month** collaboration between eRay GmbH and SRH University Heidelberg to forecast lake water quality across **4 target** indicators, chlorophyll a, turbidity, pH and dissolved oxygen, an end to end recursive time series pipeline was built over a **40 feature** space with a target lag set lag_1h, lag_24h, lag_3d, lag_7d, lag_roll_mean_24h and lag_roll_std_24h.
- **6 candidates** were benchmarked directly, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost and Prophet, with strict tree constraints max_depth 4 and learning_rate **0.05**; the decision landed on CatBoost MultiQuantile at alpha **0.05, 0.5** and **0.85**, which delivered asymmetric **80 percent** prediction intervals that hug the 0 floor and chop the top **15 percent** of summer ghost spikes.
- The September evaluation was made defensible with a **3 pass** outlier system, pH tightened from **0 to 14** down to **7.0 to 9.0**, Oct and Nov caps of **15.0** on chlorophyll a and **50.0** on turbidity, a rolling z-score at $z > 2.5$ over **48 hours**, and 5 sparse sensors plus 3 concurrent proxies phycocyanin_abs, phycocyanin_abs_comp and toc were excluded, which surfaced the honest R squared split of **0.86** on dissolved oxygen and **0.81** on pH.
- Oct and Nov gaps were reconstructed with IterativeImputer MICE, a full Memory Buffer recalculation was run across all 6 lag features, a synthetic winter canvas was generated with a 4 degree Celsius floor and **0.4** degree diurnal amplitude, and the whole thing was wrapped in an orchestrator with gate checks, ecological clips dissolved oxygen **4.0 to 18.0** and pH **6.0 to 9.0**, and a **0.003** pH per hour velocity clamp.

Technologies: Python, CatBoost, Prophet, scikit learn, MICE

Aug 2023 to Aug
2024
India

SS Engineers and Contractors

Junior Associate Software Developer

- Since SS Engineers and Contractors runs internal data dashboards, analytics platforms and employee portals used company wide, tasked with building and maintaining the frontend features teams rely on day to day, React UI components were contributed across all three internal products, which stayed in daily use by internal teams throughout the year in the role.
- For a client whose legacy AngularJS application had to run inside an existing module federation setup alongside newer React micro frontends, tasked with porting the client's screens without breaking the running product, around **8 routes** were ported from AngularJS to React inside the module federation shell, delivered over **4 months** in incremental releases and rolled out with no production outages.

Technologies: *React, module federation, Playwright, AngularJS, HTML5, CSS3*

Feb 2023 to July
2023
India

SS Engineers and Contractors

Front End Developer Intern

- During a six month internship at SS Engineers and Contractors, tasked with learning the codebase and then taking on smaller UI work in the internal data dashboards and employee portals under senior review, worked closely with senior developers through the codebase and then delivered focused UI components such as charts, filters and profile pages, each iteratively improved on code review feedback until released.

Technologies: *React, HTML5, CSS3, Git, code review workflow*

PERSONAL PROJECTS

2025

Multi-Agent RAG with LLM-as-Judge and Multilingual EN and DE Support

- For an English only hybrid RAG policy analysis system with BM25 and dense vector search over a **14 document** policy corpus that could not serve German speaking users, tasked with adding multilingual support without duplicating the corpus, embeddings and retrieval were migrated to a paraphrase multilingual MiniLM L12 v2 shared vector space, so a German query retrieves English sources and is answered end to end in German.
- Since every agent in the graph re ran language detection and produced inconsistent output languages, tasked with creating a single source of truth, a LanguageAgent was built that centralises seeded language detection with a confidence floor, propagates the output language directive to every downstream agent, and routes preprocessing to language matched spaCy pipelines with a blank multilingual fallback, logging language per chunk so retrieval and evaluation can be sliced by language.
- To measure answer quality without manual grading, an LLM as Judge JudgeAgent was implemented that scores answers **1 to 5** on **5 dimensions** (Groundedness, Relevance, Completeness, Citation Quality, Language Quality) in JSON mode at temperature 0, and self preference bias was eliminated by running the judge on a different local model, Qwen2.5 **14B**, than the generator Mistral **7B**, with a self_judged flag in every report and hard failure handling on a missing judge model so a silent fallback to self judging cannot regress unnoticed.

Technologies: *Python, LangGraph Multi Agent, Ollama with Mistral 7B and Qwen2.5 14B, Pinecone, paraphrase multilingual MiniLM L12 v2, spaCy, BM25*

EDUCATION

Apr 2025 to
present
Heidelberg

M.Sc. Data Science and Analytics

SRH University of Applied Sciences Heidelberg, GPA 1.9

2019 to 2023
Greater Noida,
India

Bachelor of Technology in Computer Science

GL Bajaj Institute of Technology and Management, CGPA 7.3 of 10

RESEARCH AND THESIS

2022 to 2023

Greater Noida,
India

Bachelor Thesis: Diabetes Prediction Using Machine Learning

GL Bajaj Institute of Technology and Management, IEEE style paper

- For a Bachelor thesis on diabetes prediction on a small clinical dataset of **768 patients**, tasked with delivering a defensible model comparison examiners could audit, a full end to end machine learning pipeline was built comparing six classifiers with **10 fold** cross validation and per model confusion matrices, producing a model comparison that held up in the defence.
- After spotting biologically impossible zero values in the source data that the original authors had overlooked, tasked with restoring data integrity before any model training, IQR based outlier handling and clean imputation were applied, bringing the dataset from silently broken to a clean training foundation for every downstream model.
- With a 65 to **35 class** imbalance that would have made accuracy a deceptively attractive headline metric, tasked with choosing an evaluation that would not hide errors in the minority class, the headline metric was switched from accuracy to ROC AUC, which exposed the real error patterns the accuracy figure had been masking and gave the thesis an honest performance comparison.
- Since the results needed to be publication ready in substance and not just submission ready, tasked with writing them up formally, an IEEE style paper was produced with an honest limitations section and suggestions for a follow up study, which the supervisor accepted as publication ready in substance.

Technologies: *Python, scikit learn, Pandas, Seaborn*

CERTIFICATES

- NVIDIA: Building LLM Applications With Prompt Engineering, issued November 2025.
- AWS Academy Graduate: AWS Academy Cloud Foundations, issued July 2025.
- Google Data Analytics: Foundations, Data, Data, Everywhere, issued April 2025 on Coursera.

ACHIEVEMENTS

- USAII Global AI Hackathon 2026: Finalist at Graduate Level, awarded by the United States Artificial Intelligence Institute for innovation, technical creativity and applied AI on real world challenges.

LANGUAGES

English	Fluent, written and spoken
German	B1, in progress toward B2
Hindi	Native language